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Inventor(s)	Khan; Atta

High temperature oxidation protection for carbon-carbon composites

Abstract

A method for forming an oxidation protection system on a composite structure may comprise: applying a ceramic layer slurry to the composite structure, wherein the ceramic layer slurry comprises aluminum and silicon in a solvent or carrier fluid; and heating the composite structure in an environment comprising nitrogen gas and oxygen gas to form a ceramic layer on the composite structure, wherein the ceramic layer comprises aluminum nitride and alumina.

Inventors:	Khan; Atta (Riverside, CA)
Applicant:	GOODRICH CORPORATION (Charlotte, NC)
Family ID:	1000008821101
Assignee:	GOODRICH CORPORATION (Charlotte, NC)
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Primary Examiner: Sahni; Vishal R

Attorney, Agent or Firm: SNELL & WILMER L.L.P.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of, claims priority to and the benefit of, U.S. patent application Ser. No. 17/527,423, filed Nov. 16, 2021 and titled “HIGH TEMPERATURE OXIDATION PROTECTION FOR CARBON-CARBON COMPOSITES,” which is incorporated by reference herein in its entirety for all purposes.

FIELD

(1) The present disclosure relates generally to composites and, more specifically, to oxidation protection systems for carbon-carbon composite structures.

BACKGROUND

(2) Oxidation protection systems for carbon-carbon composites are typically designed to minimize loss of carbon material due to oxidation at operating conditions, which include temperatures of 800° C. (1472° F.) or higher. Layers comprising ceramic materials within an oxidation protection system that are deposited onto a substrate by chemical vapor deposition (“CVD”) may provide desirable oxidation protection. However, CVD processes may be expensive and cost-prohibitive.

SUMMARY

(3) A method for forming an oxidation protection system on a composite structure is disclosed herein. The method may comprise: applying a ceramic layer slurry to the composite structure, wherein the ceramic layer slurry comprises aluminum and silicon powders in a solvent or carrier fluid; and heating the composite structure in an environment comprising nitrogen gas and oxygen gas to form a ceramic layer on the composite structure, wherein the ceramic layer comprises between 50% and 94% alumina by weight.

(4) In various embodiments, the method may further comprise: applying a sealant slurry to the composite structure, wherein the sealant slurry comprises a sealant pre-slurry composition and a sealant carrier fluid, wherein the sealant pre-slurry composition comprises a sealant phosphate glass composition; and heating the composite structure to form a sealant layer on the composite structure. The ceramic layer may be disposed between the sealant layer and the composite structure. The method may further comprise preparing the sealant slurry by combining the sealant pre-slurry composition with the sealant carrier fluid. The method may further comprise preparing the ceramic layer slurry by combining aluminum powder and silicon powder in the solvent or carrier fluid. The composite structure may be heated to a temperature between 500° C. and 1500° C. to form the ceramic layer. The composite structure may be heated to a temperature of at least 718° C. The ceramic layer may be a continuous layer comprising the alumina. The method may further comprise applying a boron compound slurry to the composite structure prior to applying the ceramic layer slurry to the composite structure. The method may further comprise allowing the boron compound slurry to dry on the composite structure to form a boron compound layer. The boron compound slurry may comprise boron carbide, and wherein the ceramic layer further comprises aluminum boron carbide. The ceramic layer slurry may further comprise silicon

oxycarbide. The ceramic layer may further comprise silicon carbide. The aluminum and the silicon, together, may comprise between 40% and 50% by weight of the ceramic layer slurry. Of the aluminum and the silicon together, the aluminum may comprise between 10% and 99% by weight, and the silicon comprises between 90% and 1% by weight. The sealant phosphate glass composition may be represented by the formula

$a(A'.\text{sub.}2O).\text{sub.}x(P.\text{sub.}2O.\text{sub.}5).\text{sub.}y1b(G.\text{sub.}fO).\text{sub.}y2c(A''O).\text{sub.}z$: A' is selected from: lithium, sodium, potassium, rubidium, cesium, and mixtures thereof; G.sub.f is selected from: boron, silicon, sulfur, germanium, arsenic, antimony, and mixtures thereof; A'' is selected from: vanadium, aluminum, tin, titanium, chromium, manganese, iron, cobalt, nickel, copper, mercury, zinc, thulium, lead, zirconium, lanthanum, cerium, praseodymium, neodymium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, actinium, thorium, uranium, yttrium, gallium, magnesium, calcium, strontium, barium, tin, bismuth, cadmium, and mixtures thereof; a is a number in the range from 1 to about 5; b is a number in the range from 0 to about 10; c is a number in the range from 0 to about 30; x is a number in the range from about 0.050 to about 0.500; y.sub.1 is a number in the range from about 0.100 to about 0.950; y.sub.2 is a number in the range from 0 to about 0.20; and z is a number in the range from about 0.01 to about 0.5; $(x+y.\text{sub.}1+y.\text{sub.}2+z)=1$; and $x < (y.\text{sub.}1+y.\text{sub.}2)$.

(5) An oxidation protection system disposed on an outer surface of a substrate is disclosed herein. The oxidation protection system may comprise: a ceramic layer comprising between 50% and 94% alumina by weight and between 5% and 50% aluminum nitride; and a sealant layer comprising a sealant pre-slurry composition comprising a sealant phosphate glass composition.

(6) In various embodiments, the ceramic layer may be a single continuous layer comprising the alumina and the aluminum nitride.

(7) An aircraft brake disk is disclosed herein. The aircraft brake disk may comprise: a carbon-carbon composite structure comprising a non-friction surface; and an oxidation protection system disposed on the non-friction surface, the oxidation protection system comprising: a ceramic layer comprising between 50% and 94% alumina by weight and between 5% and 50% aluminum nitride; and a sealant layer comprising a sealant pre-slurry composition comprising a sealant phosphate glass composition.

(8) In various embodiments, the ceramic layer is a single continuous layer comprising the alumina and the aluminum nitride.

(9) In various embodiments, an aircraft brake disk may comprise a carbon-carbon composite structure comprising a non-friction surface; and/or an oxidation protection system disposed on the non-friction surface. The oxidation protection system may comprise a ceramic layer comprising aluminum nitride; and/or a sealant layer comprising a sealant pre-slurry composition comprising a sealant phosphate glass composition. In various embodiments, the ceramic layer may further comprise silicon oxycarbide, alumina, and aluminum borocarbide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the detailed description and claims when considered in connection with the drawing figures, wherein like numerals denote like elements.

(2) FIG. 1A illustrates a cross sectional view of an aircraft wheel braking assembly, in accordance with various embodiments;

(3) FIG. 1B illustrates a partial side view of an aircraft wheel braking assembly, in accordance with various embodiments;

- (4) FIGS. 2A, 2B, and 2C illustrate methods for coating a composite structure, in accordance with various embodiments;
- (5) FIG. 3 illustrates a substrate comprising an oxidation protection system disposed thereon, in accordance with various embodiments;
- (6) FIG. 4 illustrates experimental data obtained from testing various oxidation protection systems, in accordance with various embodiments; and
- (7) FIG. 5 illustrates experimental data obtained from testing various oxidation protection systems, in accordance with various embodiments.

DETAILED DESCRIPTION

(8) The detailed description of embodiments herein makes reference to the accompanying drawings, which show embodiments by way of illustration. While these embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical, chemical, and mechanical changes may be made without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not for limitation. For example, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option. Further, any steps in a method discussed herein may be performed in any suitable order or combination.

(9) With initial reference to FIGS. 1A and 1B, aircraft wheel braking assembly **10** such as may be found on an aircraft, in accordance with various embodiments is illustrated. Aircraft wheel braking assembly may, for example, comprise a bogie axle **12**, a wheel **14** including a hub **16** and a wheel well **18**, a web **20**, a torque take-out assembly **22**, one or more torque bars **24**, a wheel rotational axis **26**, a wheel well recess **28**, an actuator **30**, multiple brake rotors **32**, multiple brake stators **34**, a pressure plate **36**, an end plate **38**, a heat shield **40**, multiple heat shield segments **42**, multiple heat shield carriers **44**, an air gap **46**, multiple torque bar bolts **48**, a torque bar pin **50**, a wheel web hole **52**, multiple heat shield fasteners **53**, multiple rotor lugs **54**, and multiple stator slots **56**. FIG. 1B illustrates a portion of aircraft wheel braking assembly **10** as viewed into wheel well **18** and wheel well recess **28**.

(10) In various embodiments, the various components of aircraft wheel braking assembly **10** may be subjected to the application of compositions and methods for protecting the components from oxidation.

(11) Brake disks (e.g., interleaved rotors **32** and stators **34**) are disposed in wheel well recess **28** of wheel well **18**. Rotors **32** are secured to torque bars **24** for rotation with wheel **14**, while stators **34** are engaged with torque take-out assembly **22**. At least one actuator **30** is operable to compress interleaved rotors **32** and stators **34** for stopping the aircraft. In this example, actuator **30** is shown as a hydraulically actuated piston, but many types of actuators are suitable, such as an electromechanical actuator. Pressure plate **36** and end plate **38** are disposed at opposite ends of the interleaved rotors **32** and stators **34**. Rotors **32** and stators **34** can comprise any material suitable for friction disks, including ceramics or carbon materials, such as a carbon/carbon composite.

(12) Through compression of interleaved rotors **32** and stators **34** between pressure plates **36** and end plate **38**, the resulting frictional contact slows rotation of wheel **14**. Torque take-out assembly **22** is secured to a stationary portion of the landing gear truck such as a bogie beam or other landing gear strut, such that torque take-out assembly **22** and stators **34** are prevented from rotating during braking of the aircraft.

(13) Carbon-carbon composites (also referred to herein as composite structures, composite substrates, and carbon-carbon composite structures, interchangeably) in the friction disks may operate as a heat sink to absorb large amounts of kinetic energy converted to heat during slowing of the aircraft. Heat shield **40** may reflect thermal energy away from wheel well **18** and back toward

rotors **32** and stators **34**. With reference to FIG. **1A**, a portion of wheel well **18** and torque bar **24** is removed to better illustrate heat shield **40** and heat shield segments **42**. With reference to FIG. **1B**, heat shield **40** is attached to wheel **14** and is concentric with wheel well **18**. Individual heat shield segments **42** may be secured in place between wheel well **18** and rotors **32** by respective heat shield carriers **44** fixed to wheel well **18**. Air gap **46** is defined annularly between heat shield segments **42** and wheel well **18**.

(14) Torque bars **24** and heat shield carriers **44** can be secured to wheel **14** using bolts or other fasteners. Torque bar bolts **48** can extend through a hole formed in a flange or other mounting surface on wheel **14**. Each torque bar **24** can optionally include at least one torque bar pin **50** at an end opposite torque bar bolts **48**, such that torque bar pin **50** can be received through wheel web hole **52** in web **20**. Heat shield segments **42** and respective heat shield carriers **44** can then be fastened to wheel well **18** by heat shield fasteners **53**.

(15) Under the operating conditions (e.g., high temperature) of aircraft wheel braking assembly **10**, carbon-carbon composites may be prone to material loss from oxidation of the carbon. For example, various carbon-carbon composite components of aircraft wheel braking assembly **10** may experience both catalytic oxidation and inherent thermal oxidation caused by heating the composite during operation. In various embodiments, composite rotors **32** and stators **34** may be heated to sufficiently high temperatures that may oxidize the carbon surfaces exposed to air. At elevated temperatures, infiltration of air and contaminants may cause internal oxidation and weakening, especially in and around rotor lugs **54** or stator slots **56** securing the friction disks to the respective torque bar **24** and torque take-out assembly **22**. Because carbon-carbon composite components of aircraft wheel braking assembly **10** may retain heat for a substantial time period after slowing the aircraft, oxygen from the ambient atmosphere may react with the carbon matrix and/or carbon fibers to accelerate material loss. Further, damage to brake components may be caused by the oxidation enlargement of cracks around fibers or enlargement of cracks in a reaction-formed porous barrier coating (e.g., a silicon-based barrier coating) applied to the carbon-carbon composite.

(16) Elements identified in severely oxidized regions of carbon-carbon composite brake components include potassium (K) and sodium (Na). These alkali contaminants may come into contact with aircraft brakes as part of cleaning or de-icing materials. Other sources include salt (e.g., NaCl) deposits left from seawater or sea spray. These and other contaminants (e.g. Ca ions, Fe ions, oxides and salts containing Fe ions and/or Ca ions, etc.) can penetrate and leave deposits in the pores of carbon-carbon composite aircraft brakes, including the substrate and any reaction-formed porous barrier coating. When such contamination occurs, the rate of carbon loss by oxidation can be increased by one to two orders of magnitude.

(17) In various embodiments, components of aircraft wheel braking assembly **10** may reach operating temperatures in the range from about 100° C. (212° F.) up to about 900° C. (1652° F.), or higher (e.g., 1093° C. (2000° F.) on a wear o'r friction surface of a brake disk). However, it will be recognized that the oxidation protection systems compositions and methods of the present disclosure may be readily adapted to many parts in this and other braking assemblies, as well as to other carbon-carbon composite structures susceptible to oxidation losses from infiltration of atmospheric oxygen and/or catalytic contaminants.

(18) In various embodiments, a method for limiting an oxidation reaction in a substrate (e.g., a composite structure) may comprise forming an oxidation protection system on the composite structure. Forming the oxidation protection system may comprise forming a ceramic layer slurry by combining aluminum and silicon in a solvent and/or carrier fluid, applying the ceramic slurry to a composite structure, and heating the composite structure to a temperature sufficient to form a ceramic layer on the composite structure. Heating the ceramic layer may include exposing the composite structure and ceramic slurry to a nitrogen gas and an oxygen gas. By exposing the ceramic slurry to oxygen gas, alumina may be formed within the ceramic layer and by exposing the ceramic slurry to nitrogen gas, aluminum nitride may be formed within the ceramic layer. In this

regard, the heating process may facilitate a combination of aluminum nitride and alumina within the oxidation protection system as described further herein.

(19) In various embodiments, forming the oxidation protection system may further comprise forming a sealant pre-slurry composition, comprising a sealant phosphate glass composition (in the form of a glass frit, powder, or other suitable pulverized form), with a carrier fluid (such as, for example, water), applying the sealant slurry to a composite structure, and heating the composite structure to a temperature sufficient to dry the carrier fluid and form an oxidation protection coating on the composite structure, which in various embodiments may be referred to a sealant layer. In various embodiments, the sealant pre-slurry composition of the sealant slurry may comprise additives, such as, for example, ammonium dihydrogen phosphate and/or aluminum orthophosphate, among others, to improve hydrolytic stability and/or to increase the composite structure's resistance to oxidation, thereby tending to reduce mass loss of composite structure.

(20) With initial reference to FIGS. 1A and 2A, a method **200** for coating a composite structure in accordance with various embodiments is illustrated. Method **200** may, for example, comprise applying an oxidation protection system to non-wearing surfaces of carbon-carbon composite brake components, such as non-wear surfaces **45** and/or rotor lugs **54**. Non-wear surfaces (e.g., non-wear surface **45**, as labeled in FIG. 1A) simply reference an exemplary non-wear surface on a brake disk (i.e., a non-friction surface that does not contribute to brake function by having friction with another component, such as another brake disk), but non-wear surfaces similar to non-wear surface **45** may be present on any brake disks (e.g., rotors **32**, stators **34**, pressure plate **36**, end plate **38**, or the like). In various embodiments, method **200** may be used on the back face of pressure plate **36** and/or end plate **38**, an inner diameter (ID) surface of stators **34** including stator slots **56**, as well as outer diameter (OD) surfaces of rotors **32** including lugs **54**. The oxidation inhibiting composition of method **200** may be applied to preselected regions of a carbon-carbon composite structure that may be otherwise susceptible to oxidation. For example, aircraft brake disks may have the oxidation inhibiting composition applied on or proximate stator slots **56**, rotor lugs **54**, and/or non-wear surface **45**.

(21) In various embodiments, method **200** may comprise forming a ceramic layer slurry (step **210**). The ceramic layer slurry may comprise aluminum and silicon. The aluminum and silicon may be elemental aluminum and silicon and/or alloys or compounds comprising aluminum and silicon atoms or ions. The aluminum and/or silicon may be in powder form. The aluminum and silicon may be added to a solvent and/or carrier fluid. The solvent and/or carrier fluid may comprise any suitable fluid, such as water, methyltrimethoxysilane ("MTMS"), hexane, cyclohexane, polyvinyl alcohol (PVA), isopropyl alcohol (or other alcohol), any combination of the foregoing, and/or the like. In various embodiments, the aluminum and silicon may be mixed into the solvent and/or carrier fluid via any suitable method, such as stirring or tumbling. The aluminum and silicon may be mixed into the solvent and/or carrier fluid for any suitable duration. For example, such mixing may take place for under an hour, over an hour, or for multiple hours, between two and ten hours. In various embodiments the aluminum and silicon may be mixed into the solvent and/or carrier fluid for between two and four hours, for about two hours, or about three hours (the term "about" as used in this context means plus or minus 30 minutes).

(22) In various embodiments, of the aluminum and silicon (collectively, the "solids"), the aluminum may comprise between 10% and 100% by weight, between 10% and 99% by weight, or between 10% and 90% by weight, and the silicon may comprise between 90% and 0% by weight, between 90% and 1% by weight, or between 90% and 10% by weight. In various embodiments, of the aluminum and silicon, the aluminum may comprise about 90% by weight, and the silicon may comprise about 10% by weight (the term "about" in this context means plus or minus 10 weight percent).

(23) In various embodiments, regarding the compositional make up the ceramic layer slurry, the aluminum/silicon solids may comprise between 15% and 80% by weight of the ceramic layer

slurry, between 30% and 70% by weight of the ceramic layer slurry, between 40% and 60% by weight of the ceramic layer slurry, or between 40% and 50% by weight of the ceramic layer slurry. In various embodiments, the aluminum/silicon solids may comprise about 40% or about 45% by weight of the ceramic layer slurry (the term “about” in this context means plus or minus 10 weight percent). In various embodiments, the solvent(s) and/or carrier fluid(s) may comprise between 15% and 80% by weight, or between 15% and 90% by weight of the ceramic layer slurry, or about 40% by weight of the ceramic layer slurry (the term “about” in this context means plus or minus 10 weight percent).

(24) In various embodiments, the ceramic layer slurry may further comprise a binder (but in further embodiments, the ceramic layer slurry may not comprise a binder). The binder may comprise any suitable compound, such as silicon oxycarbide, silicon oxycarbide generating sol, silicon oxycarbide pre-ceramic polymer, and/or polyvinyl alcohol. In various embodiments, the binder may comprise between 0.2% and 15% by weight of the ceramic layer slurry, between 5% and 15% by weight of the ceramic layer slurry, or about 10% or about 12% by weight of the ceramic layer slurry (the term “about” in this context means plus or minus 5 weight percent).

(25) In various embodiments, method **200** further comprises applying the ceramic layer slurry to a composite structure (step **220**). Applying the ceramic layer slurry may comprise, for example, spraying or brushing the ceramic layer slurry to an outer surface of the composite structure (e.g., a non-wear or non-friction surface). Any suitable manner of applying the ceramic layer slurry to the composite structure is within the scope of the present disclosure. As referenced herein, the composite structure may refer to a carbon-carbon composite structure.

(26) In various embodiments, method **200** may further comprise a step **230** of heating the composite structure to form a ceramic layer. The composite structure may be heated (e.g., dried or baked) at a temperature in the range from about 500° C. (932° F.) to about 1500° C. (2732° F.). In various embodiments, the composite structure may be heated to a temperature in a range from about 500° C. (932° F.) to about 1200° C. (2192° F.), or between about 600° C. (1112° F.) to about 1100° C. (2012° F.). In various embodiments, the composite structure may be heated to, or at least to, about 1000° C. (1832° F.), or to, or at least to, about 600° C. (1112° F.), which may depend on the environment in which the composite structure is being heated (the term “about” in this context means plus or minus 100° C. (212° F.)). Step **230** may, for example, comprise heating the composite structure for a period between about 0.5 hour and about eight hours, or about three hours or about four hours (the term “about” in this context means plus or minus one hour). The ceramic layer may also be referred to as a coating. The temperature rise may be controlled at a rate that removes water without boiling, and provides temperature uniformity throughout the composite structure.

(27) In various embodiments, step **230** may be performed in an environment comprising nitrogen. The nitrogen may be comprised in a gas, such as nitrogen gas (N.sub.2) or ammonia gas (NH.sub.3). During the heating, the nitrogen in the surrounding environment may react with the aluminum in the ceramic layer slurry to form aluminum nitride. Therefore, the ceramic layer may comprise aluminum nitride. At elevated temperatures, the silicon may also react with the nitrogen in the surrounding environment, forming silicon nitride. Therefore, the ceramic layer may comprise silicon nitride. Also, in embodiments including a silicon oxycarbide binder in the ceramic layer slurry, the aluminum may react with the silicon oxycarbide to form alumina (Al.sub.2O.sub.3) and silicon carbide. The reaction between the aluminum and silicon oxycarbide within the ceramic layer slurry (to form alumina) aids in binding the compounds of the alumina ceramic layer slurry and the resulting ceramic layer to itself, thus strengthening the ceramic layer. Also, the reaction between the aluminum, silicon, silicon oxycarbide, and/or the carbon of the composite structure (forming silicon carbide) aids in binding the ceramic layer to the composite structure.

(28) In various embodiments, the environment for step **230** may further comprise oxygen. The oxygen may be comprised in a gas, such as oxygen gas (O.sub.2). During the heating, the oxygen

gas may comprise between 0.1% and 10% by volume, or between 0.1% and 5% by volume. The nitrogen gas may comprise between 99.9% and 90% by volume, or between 99.9% and 95% by volume. In various embodiments, a furnace with reduced sealant capability may be utilized. In this regard, oxygen within the environment for step **230** may be oxygen leaked into the environment from an external environment. Thus, significantly less expensive furnaces may be utilized, resulting in reduced costs and producing a more effective oxidation protection system relative to typical systems, in accordance with various embodiments. In various embodiments, by exposing the slurry to nitrogen gas and oxygen gas, a continuous layer of silicon oxycarbide and alumina may be formed within a single layer. In various embodiments, the ceramic layer formed from method **200** may comprise between 50% and 94% alumina, or between 60% and 90%, or between 70% and 90%, or approximately 80% by weight. In various embodiments, the ceramic layer formed from method **200** may comprise between 5% and 50% silicon oxycarbide and pure silicon between 1% and 30% by weight.

(29) In various embodiments, in response to being heated in an environment comprising ammonia gas, the substrate may be heated to a temperature of at least 600° C. (1112° F.) or about 600° C. (1112° F.) (the term “about” in this context means plus or minus 100° C. (212° F.)). At such temperatures, the ammonia gas may react with the aluminum in the ceramic layer slurry to form aluminum nitride comprised in the resulting ceramic layer. In such an environment comprising ammonia, silicon may react with ammonia to form silicon nitride at about 900° C. (1652° F.). Thus, in various embodiments, in an ammonia environment, the substrate may be heated to at least 900° C. (1652° F.), or to about 900° C. (1652° F.), to form aluminum nitride and silicon nitride.

(30) In various embodiments, in response to being heated in an environment comprising nitrogen gas (which is less reactive than ammonia), the substrate may be heated to a temperature of at least 900° C. (1652° F.) or about 900° C. (1652° F.) (the term “about” in this context means plus or minus 100° C. (212° F.)). At such temperatures, the nitrogen gas may react with the aluminum in the ceramic layer slurry to form aluminum nitride comprised in the resulting ceramic layer. In such an environment comprising nitrogen gas, silicon may react with nitrogen to form silicon nitride at about 1100° C. (2012° F.). Thus, in various embodiments, in a nitrogen gas environment, the substrate may be heated to at least 1100° C. (2012° F.), or to about 1100° C. (2012° F.), to form aluminum nitride and silicon nitride.

(31) The pressure during heating of the substrate may be atmospheric pressure, or between 5 pascals and 150,000 pascals, or between 50 and 100,000 pascals, or between 100 and 50,000 pascals.

(32) In various embodiments and with reference now to FIG. 2B, method **300**, which comprises steps also found in method **200**, may further comprise applying at least one of a pretreating composition or a barrier coating (step **215**) prior to applying the ceramic layer slurry. Step **215** may, for example, comprise applying a first pretreating composition to an outer surface of a composite structure, such as a component of aircraft wheel braking assembly **10**. In various embodiments, the first pretreating composition comprises an aluminum oxide in water. For example, the aluminum oxide may comprise an additive, such as a nanoparticle dispersion of aluminum oxide (for example, NanoBYK-3600®, sold by BYK Additives & Instruments). The first pretreating composition may further comprise a surfactant or a wetting agent. The composite structure may be porous, allowing the pretreating composition to penetrate at least a portion of the pores of the composite structure.

(33) In various embodiments, after applying the first pretreating composition, the component may be heated to remove water and fix the aluminum oxide in place. For example, the component may be heated between about 100° C. (212° F.) and 200° C. (392° F.), and further, between 100° C. (212° F.) and 150° C. (302° F.).

(34) Step **215** may further comprise applying a second pretreating composition. In various embodiments, the second pretreating composition may comprise a phosphoric acid and an aluminum phosphate, aluminum hydroxide, and/or aluminum oxide. The second pretreating

composition may further comprise, for example, a second metal salt such as a magnesium salt. In various embodiments, the aluminum to phosphorus molar ratio of the aluminum phosphate is 1 to 3 or less. Further, the second pretreating composition may also comprise a surfactant or a wetting agent. In various embodiments, the second pretreating composition is applied to the composite structure atop the first pretreating composition. The composite structure may then, for example, be heated. In various embodiments, the composite structure may be heated between about 600° C. (1112° F.) and about 800° C. (1472° F.), and further, between about 650° C. (1202° F.) and 750° C. (1382° F.).

(35) In various embodiments, step **215** may comprise applying a barrier coating to an outer surface of a composite structure (alternatively or in addition to the pretreating compositions), such as a component of aircraft wheel braking assembly **10**, prior to application of the ceramic layer slurry. In various embodiments, the barrier coating composition may comprise carbides or nitrides, including at least one of a boron nitride, silicon carbide, titanium carbide, boron carbide, silicon oxycarbide, and silicon nitride. In various embodiments, the barrier coating may be a boron compound layer formed from a boron compound slurry. The boron compound slurry may comprise a boron compound comprising boron, such as boron, boron carbide, boron nitride, titanium boride, and a solvent and/or carrier fluid (e.g., water, isopropyl alcohol or another alcohol, hexanes, cyclohexane, and/or the like). In embodiments in which the boron compound comprises boron carbide, the boron carbide may comprise a powder comprising granules having sizes ranging from 0.4 micrometer (1.575×10^{-5} inch) to 100 micrometers (3.94×10^{-3} inch) in size. A boron compound slurry may comprise between 10% and 60% by weight boron compound (e.g., boron carbide), and the remainder comprising solvent and/or carrier fluid. A boron compound slurry may further comprise a dispersant, which may be any suitable dispersant. For example, the dispersant may comprise aluminum oxide (for example, NanoBYK-3600®, sold by BYK Additives & Instruments). A boron compound slurry may comprise less than 1% by weight dispersant.

(36) The boron compound slurry may be applied to the composite structure in any suitable manner (spraying, brushing, etc.) and then dried to form a boron compound layer. In various embodiments, the ceramic layer slurry may be applied to the boron compound layer. In response to heating the composite structure to form the ceramic layer (step **230**), the boron compound from the boron compound layer (e.g., boron carbide) may react with the aluminum comprised in the ceramic layer slurry to form aluminum boro-carbide. In various embodiments, the ceramic layer resulting from step **230** may comprise aluminum boro-carbide. In various embodiments, the ceramic layer may at least partially comprise the boron compound layer, or there may be overlap between the ceramic layer and the boron compound layer.

(37) In various embodiments and with reference now to FIG. 2C, method **400** may further comprise a step **240**, of forming a sealant slurry by combining a sealant pre-slurry composition, which may comprise a sealant phosphate glass composition in glass frit or powder form, with a carrier fluid (such as, for example, water). In various embodiments, the sealant pre-slurry composition may further comprise ammonium dihydrogen phosphate (“ADHP”) and/or aluminum orthophosphate. The sealant slurry may be applied to the composite structure (step **245**), for example, by spraying or brushing the sealant slurry on to an outer surface of the ceramic layer. Any suitable manner of applying the sealant slurry to the ceramic layer and/or composite structure is within the scope of the present disclosure (e.g., the application methods described in relation to step **220**).

(38) In various embodiments, the sealant phosphate glass composition may comprise phosphate glass in the form of a glass frit, powder, or other suitable pulverized and/or ground form, with a carrier fluid (such as, for example, water). The sealant phosphate glass composition may comprise and/or be combined with one or more alkali metal glass modifiers, one or more glass network modifiers and/or one or more additional glass formers. In various embodiments, boron oxide or a precursor may optionally be combined with the $P_{2.5}O_{5}$ mixture to form a borophosphate glass, which has improved self-healing properties at the operating temperatures typically seen in

aircraft braking assemblies. In various embodiments, the phosphate glass and/or borophosphate glass may be characterized by the absence of an oxide of silicon. Further, the ratio of P_2O_5 to metal oxide in the fused glass may be in the range from about 0.25 to about 5 by weight.

(39) Potential alkali metal glass modifiers may be selected from oxides of lithium, sodium, potassium, rubidium, cesium, and mixtures thereof. In various embodiments, the glass modifier may be an oxide of lithium, sodium, potassium, or mixtures thereof. These or other glass modifiers may function as fluxing agents. Additional glass formers can include oxides of boron, silicon, sulfur, germanium, arsenic, antimony, and mixtures thereof.

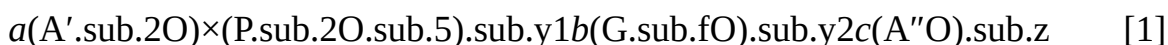
(40) Suitable glass network modifiers include oxides of vanadium, aluminum, tin, titanium, chromium, manganese, iron, cobalt, nickel, copper, mercury, zinc, thulium, lead, zirconium, lanthanum, cerium, praseodymium, neodymium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, actinium, thorium, uranium, yttrium, gallium, magnesium, calcium, strontium, barium, tin, bismuth, cadmium, and mixtures thereof.

(41) The sealant phosphate glass composition may be prepared by combining the above ingredients and heating them to a fusion temperature. In various embodiments, depending on the particular combination of elements, the fusion temperature may be in the range from about 700° C. (1292° F.) to about 1500° C. (2732° F.). The resultant melt may then be cooled and pulverized and/or ground to form a glass frit or powder. In various embodiments, the sealant phosphate glass composition may be annealed to a rigid, friable state prior to being pulverized. Glass transition temperature (T_g), glass softening temperature (T_s) and glass melting temperature (T_m) may be increased by increasing refinement time and/or temperature. Before fusion, the sealant phosphate glass composition comprises from about 20 mol % to about 80 mol % of P_2O_5 . In various embodiments, the sealant phosphate glass composition comprises from about 30 mol % to about 70 mol % P_2O_5 , or precursor thereof. In various embodiments, the sealant phosphate glass composition comprises from about 40 to about 60 mol % of P_2O_5 . In this context, the term “about” means plus or minus 5 mol %.

(42) The sealant phosphate glass composition may comprise, or be combined with, from about 5 mol % to about 50 mol % of the alkali metal oxide. In various embodiments, the sealant phosphate glass composition may comprise, or be combined with, from about 10 mol % to about 40 mol % of the alkali metal oxide. Further, the sealant phosphate glass composition may comprise, or be combined with, from about 15 to about 30 mol % of the alkali metal oxide or one or more precursors thereof. In various embodiments, the sealant phosphate glass composition may comprise, or be combined with, from about 0.5 mol % to about 50 mol % of one or more of the above-indicated glass formers. The sealant phosphate glass composition may comprise, or be combined with, about 5 to about 20 mol % of one or more of the above-indicated glass formers. As used herein, mol % is defined as the number of moles of a constituent per the total moles of the solution.

(43) In various embodiments, the sealant phosphate glass composition may comprise, or be combined with, from about 0.5 mol % to about 40 mol % of one or more of the above-indicated glass network modifiers. The sealant phosphate glass composition may comprise, or be combined with, from about 2.0 mol % to about 25 mol % of one or more of the above-indicated glass network modifiers.

(44) In various embodiments, the sealant phosphate glass composition may be represented by the formula:



(45) In Formula 1, A' is selected from: lithium, sodium, potassium, rubidium, cesium, and mixtures thereof; G.sub.f is selected from: boron, silicon, sulfur, germanium, arsenic, antimony, bismuth, and mixtures thereof; A'' is selected from: vanadium, aluminum, tin, titanium, chromium, manganese, iron, cobalt, nickel, copper, mercury, zinc, thulium, lead, zirconium, lanthanum,

cerium, praseodymium, neodymium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, actinium, thorium, uranium, yttrium, gallium, magnesium, calcium, strontium, barium, tin, bismuth, cadmium, and mixtures thereof; a is a number in the range from 1 to about 5; b is a number in the range from 0 to about 10; c is a number in the range from 0 to about 30; x is a number in the range from about 0.050 to about 0.500; y.sub.1 is a number in the range from about 0.100 to about 0.950; y.sub.2 is a number in the range from 0 to about 0.20; and z is a number in the range from about 0.01 to about 0.5; $(x+y_{\text{sub.1}}+y_{\text{sub.2}}+z)=1$; and $x < (y_{\text{sub.1}}+y_{\text{sub.2}})$. The sealant phosphate glass composition may be formulated to balance the reactivity, durability and flow of the resulting glass base layer for optimal performance. As used in this context, the term “about” means plus or minus ten percent of the respective value.

(46) The sealant slurry may comprise any suitable weight percentage sealant phosphate glass composition. For example, the sealant slurry may comprise between 20% and 50% by weight sealant phosphate glass composition, between 20% and 40% by weight sealant phosphate glass composition, between 20% and 30% by weight sealant phosphate glass composition, and/or between 30% and 40% by weight sealant phosphate glass composition. The sealant pre-slurry composition (and/or the resulting sealant layer, discussed in association with step **250**) may comprise any suitable weight percentage sealant phosphate glass composition. For example, the sealant pre-slurry composition may comprise between 50% and 95% by weight sealant phosphate glass composition, between 60% and 90% by weight sealant phosphate glass composition, and/or between 70% and 80% by weight sealant phosphate glass composition.

(47) In various embodiments, method **200** may further comprise a step **250** of heating the composite structure to form a sealant layer comprising phosphate glass over the ceramic layer. The composite structure may be heated (e.g., dried or baked) at a temperature in the range from about 200° C. (292° F.) to about 1000° C. (1832° F.). In various embodiments, the composite structure is heated to a temperature in a range from about 600° C. (1112° F.) to about 1000° C. (1832° F.), or between about 200° C. (292° F.) to about 900° C. (1652° F.), or further, between about 400° C. (752° F.) to about 850° C. (1562° F.). Step **250** may, for example, comprise heating the composite structure for a period between about 0.5 hour and about 8 hours, wherein the term “about” in this context only means plus or minus 0.25 hours. The sealant layer may also be referred to as a coating.

(48) In various embodiments, the composite structure may be heated to a first, lower temperature (for example, about 30° C. (86° F.) to about 400° C. (752° F.)) to bake or dry the sealant layer at a controlled depth. A second, higher temperature (for example, about 300° C. (572° F.) to about 1000° C. (1832° F.)) may then be used to melt the phosphate glass composition, creating a substantially uniform sealant layer over the ceramic layer. The duration of each heating step can be determined as a fraction of the overall heating time and can range from about 10% to about 50%, wherein the term “about” in this context only means plus or minus 5%. In various embodiments, the duration of the lower temperature heating step(s) can range from about 20% to about 40% of the overall heating time, wherein the term “about” in this context only means plus or minus 5%. The lower temperature step(s) may occupy a larger fraction of the overall heating time, for example, to provide relatively slow heating up to and through the first lower temperature. The exact heating profile will depend on a combination of the first temperature and desired depth of the drying portion.

(49) Step **250** may be performed in an inert environment, such as under a blanket of inert gas or less reactive gas (e.g., nitrogen, argon, other noble gases, and the like). For example, a composite structure may be pretreated or warmed prior to application of the sealant slurry to aid in the penetration of the sealant slurry. Step **250** may be for a period of about 2 hours at a temperature of about 600° C. (1112° F.) to about 900° C. (1652° F.), wherein the term “about” in this context only means plus or minus 50° C. The composite structure and the sealant slurry may then be dried or baked in a non-oxidizing, inert or less reactive atmosphere, e.g., noble gasses and/or nitrogen

(N.sub.2), to optimize the retention of the sealant pre-slurry composition of the sealant slurry and resulting sealant layer filling any pores or cracks in the ceramic layer. This retention may, for example, be improved by heating the composite structure to about 200° C. (392° F.) and maintaining the temperature for about 1 hour before heating the carbon-carbon composite to a temperature in the range described above. The temperature rise may be controlled at a rate that removes water without boiling, and provides temperature uniformity throughout the composite structure.

(50) At elevated operation temperatures of aircraft brake disks (e.g., 760° C. (1400° F.) and above), the ceramic layer may crack, which causes a risk of oxygen penetrating through the oxidation protection system and allowing oxidation of the composite structure. At such elevated temperatures, the sealant layer may melt and/or flow, allowing the sealant layer to fill any cracks that may form in the ceramic layer, thus, sealant the ceramic layer and the oxidation protection system.

(51) In various embodiments, the sealant slurry may comprise an additional metal salt. The cation of the additional metal salt may be multivalent. The metal may be an alkaline earth metal or a transition metal. In various embodiments, the metal may be an alkali metal. The multivalent cation may be derived from a non-metallic element such as boron. The term “metal” is used herein to include multivalent elements such as boron that are technically non-metallic. The metal of the additional metal salt may be an alkaline earth metal such as calcium, magnesium, strontium, barium, or a mixture of two or more thereof. The metal for the additional metal salt may be iron, manganese, tin, zinc, or a mixture of two or more thereof. The anion for the additional metal salt may be an inorganic anion such as a phosphate, halide, sulfate or nitrate, or an organic anion such as acetate. In various embodiments, the additional metal salt may be an alkaline earth metal salt such as an alkaline earth metal phosphate. In various embodiments, the additional metal salt may be a magnesium salt such as magnesium phosphate. In various embodiments, the additional metal salt may be an alkaline earth metal nitrate, an alkaline earth metal halide, an alkaline earth metal sulfate, an alkaline earth metal acetate, or a mixture of two or more thereof. In various embodiments, the additional metal salt may be magnesium nitrate, magnesium halide, magnesium sulfate, or a mixture of two or more thereof. In various embodiments, the additional metal salt may comprise: (i) magnesium phosphate; and (ii) a magnesium nitrate, magnesium halide, magnesium sulfate, or a mixture of two or more thereof.

(52) The additional metal salt may be selected with reference to its compatibility with other ingredients in the sealant slurry. Compatibility may include metal phosphates that do not precipitate, flocculate, agglomerate, react to form undesirable species, or settle out prior to application of the sealant slurry to the carbon-carbon composite. The phosphates may be monobasic (H.sub.2PO.sub.4.sup.-), dibasic (HPO.sub.4.sup.-2), or tribasic (PO.sub.4.sup.-3). The phosphates may be hydrated. Examples of alkaline earth metal phosphates that may be used include calcium hydrogen phosphate (calcium phosphate, dibasic), calcium phosphate tribasic octahydrate, magnesium hydrogen phosphate (magnesium phosphate, dibasic), magnesium phosphate tribasic octahydrate, strontium hydrogen phosphate (strontium phosphate, dibasic), strontium phosphate tribasic octahydrate and barium phosphate.

(53) In one embodiment, a chemical equivalent of the additional metal salt may be used as the additional metal salt. Chemical equivalents include compounds that yield an equivalent (in this instance, an equivalent of the additional metal salt) in response to an outside stimulus such as, temperature, hydration, or dehydration. For example, equivalents of alkaline earth metal phosphates may include alkaline earth metal pyrophosphates, hypophosphates, hypophosphites and orthophosphites. Equivalent compounds include magnesium and barium pyrophosphate, magnesium and barium orthophosphate, magnesium and barium hypophosphate, magnesium and barium hypophosphite, and magnesium and barium orthophosphite.

(54) While not wishing to be bound by theory, it is believed that the addition of multivalent cations,

such as alkaline earth metals, transition metals and nonmetallic elements such as boron, to the sealant slurry enhances the hydrolytic stability of the metal-phosphate network. In general, the hydrolytic stability of the metal-phosphate network increases as the metal content increases, however a change from one metallic element to another may influence oxidation inhibition to a greater extent than a variation in the metal-phosphate ratio. The solubility of the phosphate compounds may be influenced by the nature of the cation associated with the phosphate anion. For example, phosphates incorporating monovalent cations such as sodium orthophosphate or phosphoric acid (hydrogen cations) are very soluble in water, while (tri)barium orthophosphate is insoluble. Phosphoric acids can be condensed to form networks but such compounds tend to remain hydrolytically unstable. Generally, it is believed that the multivalent cations link phosphate anions creating a phosphate network with reduced solubility. Another factor that may influence hydrolytic stability is the presence of —P—O—H groups in the condensed phosphate product formed from the sealant slurry during thermal treatment. The sealant slurry may be formulated to minimize concentration of these species and any subsequent hydrolytic instability. Whereas increasing the metal content may enhance the hydrolytic stability of the sealant slurry, it may be desirable to strike a balance between composition stability and effectiveness as an oxidation inhibitor.

(55) In various embodiments, the additional metal salt may be present in the sealant slurry at a concentration in the range from about 0.5 weight percent to about 30 weight percent, and in various embodiments from about 0.5 weight percent to about 25 weight percent, and in various embodiments from about 5 weight percent to about 20 weight percent. In various embodiments, a combination of two or more additional metal salts may be present at a concentration in the range from about 10 weight percent to about 30 weight percent, and in various embodiments from about 12 weight percent to about 20 weight percent.

(56) FIG. 3 depicts a substrate **305** (e.g., a composite structure) with an oxidation protection system **350** disposed thereon (e.g., on a non-wear or non-friction surface). In accordance with various embodiments, the oxidation protection system on a substrate may comprise a ceramic layer (e.g., layer **320**). The ceramic layer may be directly disposed on and in contact with the substrate. In various embodiments, an oxidation protection system may comprise a layer(s) between the substrate **305** and the ceramic layer, for example, layer **310**. Layer **310** may comprise a pretreating layer(s) or a barrier coating (e.g., a boron compound layer), as discussed herein. In various embodiments, an oxidation protection system may comprise a sealant layer (e.g., layer **330**). The sealant layer may be disposed on, and directly contact, the ceramic layer, such that the ceramic layer is disposed between the sealant slurry and the substrate.

(57) TABLE 1 illustrates a sealant slurry prepared in accordance with the embodiments discussed herein. Each numerical value in TABLE 1 is the number of grams of the particular substance added to the slurry.

(58) TABLE-US-00001 TABLE 1 Example >> A H.sub.2O 52.4 Ammonium dihydrogen phosphate (ADHP) 11.33 Glass frit 34 Aluminum orthophosphate (o-AlPO.sub.4) 2.27

(59) As illustrated in TABLE 1, an oxidation protection system sealant slurry (slurry A) comprises a pre-slurry composition, comprising phosphate glass composition glass frit and various additives such as aluminum orthophosphate, and/or ammonium dihydrogen phosphate, in a carrier fluid (i.e., water), was prepared. Slurry A may be a suitable sealant slurry which will serve as a sealant layer after heating (such as during step **250**).

(60) With combined reference to TABLE 1, FIG. 4, and FIG. 5, TABLE 1, plot **400** in FIG. 4, and plot **500** in FIG. 5 may allow evaluation of oxidation protection systems comprising a ceramic layer and a sealant layer versus oxidation protection systems comprising and a sealant layer without a ceramic layer. Data sets **402** and **406** represents an oxidation protection system comprising a ceramic layer and a sealant layer, while data sets **404** and **408** represents an oxidation protection systems comprising a sealant layer (resulting from slurry A) without a ceramic layer. Percent weight loss is shown on the y-axis and exposure time is shown on the x-axis of the graph depicted

in FIGS. 4 and 5.

(61) The oxidation protection system represented by data sets **404** and **408** were prepared by applying slurry A to carbon-carbon composite coupons and heated to form a sealant slurry, in accordance with embodiments discussed herein. Slurry A for data sets **404** and **408** are not hydrated (i.e., dry). Example 1, represented by data sets **402** and **406**, were prepared by: (1) applying a boron compound slurry comprising 14 grams of boron carbide, 86 grams of water, and 0.2 grams of dispersant to a carbon-carbon composite coupon, by dipping the coupon in the boron compound slurry; (2) air-drying the boron compound slurry to form a boron compound layer on the coupon; (3) preparing a ceramic layer slurry comprising 16.76 grams of MTMS, 41.57 grams of water, 4.32 grams of sub-micron silicon powder, and 37.35 grams of 30-micron aluminum powder; (4) applying the ceramic layer slurry to the boron compound layer by dipping the coupon in the ceramic layer slurry; (5) air-drying the ceramic layer slurry; (6) heating the coupon at 950° C. (1742° F.) for about three hours under an environment comprising nitrogen gas and oxygen gas to form the ceramic layer comprising at least aluminum nitride and alumina; (7) applying slurry A to the ceramic layer; and (8) heating the coupon at 870° C. (1598° F.) to melt the phosphate glass in the sealant slurry and form the sealant layer. The coupons were then heated in accordance with the time shown on the x-axis and temperatures shown in the header of FIGS. 4 and 5.

(62) As can be seen in FIGS. 4 and 5, the oxidation protection systems having the ceramic layer and the sealant layer (data sets **402** and **406**) resulted in drastically less weight loss of the composite structure than the oxidation protection systems having a sealant layer with no ceramic layer (data sets **404** and **408**). That is, oxidation protection systems having the ceramic layer and the sealant layer (data sets **402** and **406**) resulted in about 0.03% weight gain at 871° C. (1600° F.) (FIG. 4) and 0.06% weight gain at 677° C. (1250° F.) (FIG. 5) of the substrate at the end of the test, while the oxidation protection system having a sealant layer with no ceramic layer (data sets **404** and **408**) resulted in about 65% and 1% weight loss, respectively. Thus, oxidation protection systems having the ceramic layer and the sealant layer (data sets **402** and **406**) provided significantly greater oxidation protection than the oxidation protection system having the sealant layer only (data sets **404** and **408**). These results indicate that the oxidation protection systems comprising the ceramic layer (comprising aluminum oxide (major component), aluminum nitride, silicon, silicon nitride, aluminum boron carbide, and/or silicon carbide), along with the sealant layer disposed thereon, creates a strong barrier which stops oxygen from passing therethrough and causing oxidation of the underlying substrate (especially at elevated temperatures of 1600° F.). The economical preparation and application of the involved slurries, the significantly reduced cost in furnace systems, along with the positive and improved oxidation protection results, indicate that the systems and methods discussed herein provide effective oxidation protection, in accordance with various embodiments.

(63) Benefits and other advantages have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and

B and C.

(64) Systems, methods, and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic.

Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(65) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Claims

1. An oxidation protection system disposed on an outer surface of a substrate, comprising: a ceramic layer consisting of between 50% and 94% alumina by weight between 5% and 50% aluminum nitride, pure silicon between 1% and 30% by weight, and optionally a binder having a remaining weight, wherein the ceramic layer is formed by: preparing a ceramic layer slurry by combining only an aluminum powder and a silicon powder in a solvent or a carrier fluid, wherein the aluminum powder and the silicon powder comprise between 40% and 60% by weight of the ceramic layer slurry; applying the ceramic layer slurry to a composite structure, wherein the ceramic layer slurry comprises the aluminum powder and the silicon powder in the solvent or the carrier fluid; and heating the composite structure in an environment comprising nitrogen gas and oxygen gas to form the ceramic layer on the composite structure; and a sealant layer comprising a sealant pre-slurry composition comprising a sealant phosphate glass composition.
2. The oxidation protection system of claim 1, wherein the ceramic layer is a single continuous layer comprising the alumina and the aluminum nitride.
3. An aircraft brake disk, comprising: a carbon-carbon composite structure comprising a non-friction surface; and an oxidation protection system disposed on the non-friction surface, the oxidation protection system comprising: a ceramic layer consisting of between 50% and 94% alumina by weight, between 5% and 50% aluminum nitride, pure silicon between 1% and 30% by weight, and optionally a binder having a remaining weight, wherein the ceramic layer is formed by: preparing a ceramic layer slurry by combining only an aluminum powder and a silicon powder in a solvent or a carrier fluid, wherein the aluminum powder and the silicon powder comprise between 40% and 60% by weight of the ceramic layer slurry; applying the ceramic layer slurry to a composite structure, wherein the ceramic layer slurry comprises the aluminum powder and the silicon powder in the solvent or the carrier fluid; and heating the composite structure in an environment comprising nitrogen gas and oxygen gas to form the ceramic layer on the composite structure; and a sealant layer comprising a sealant pre-slurry composition comprising a sealant phosphate glass composition.
4. The aircraft brake disk of claim 3, wherein the ceramic layer is a single continuous layer comprising the alumina and the aluminum nitride.

5. The oxidation protection system of claim 1, wherein sealant layer is formed by: preparing a sealant slurry by combining a sealant pre-slurry composition with a carrier fluid; applying the sealant slurry to an outer surface of the ceramic layer; and heating the composite structure to form the sealant layer over the ceramic layer.
 6. The oxidation protection system of claim 5, wherein the sealant pre-slurry composition comprises a sealant phosphate glass composition in glass frit form.
 7. The oxidation protection system of claim 5, wherein the sealant pre-slurry composition comprises a sealant phosphate glass composition in powder form.
 8. The oxidation protection system of claim 5, wherein the sealant pre-slurry composition further comprises ammonium dihydrogen phosphate.
 9. The oxidation protection system of claim 5, wherein the sealant pre-slurry composition further comprises aluminum orthophosphate.
 10. The oxidation protection system of claim 5, wherein the carrier fluid is water.
 11. The aircraft brake disk of claim 3, wherein sealant layer is formed by: preparing a sealant slurry by combining a sealant pre-slurry composition with a carrier fluid; applying the sealant slurry to an outer surface of the ceramic layer; and heating the composite structure to form the sealant layer over the ceramic layer.
 12. The aircraft brake disk of claim 11, wherein the sealant pre-slurry composition comprises a sealant phosphate glass composition in glass frit form.
 13. The aircraft brake disk of claim 11, wherein the sealant pre-slurry composition comprises a sealant phosphate glass composition in powder form.
 14. The aircraft brake disk of claim 11, wherein the sealant pre-slurry composition further comprises ammonium dihydrogen phosphate.
 15. The aircraft brake disk of claim 11, wherein the sealant pre-slurry composition further comprises aluminum orthophosphate.
 16. The aircraft brake disk of claim 11, wherein the carrier fluid is water.
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